

Adversarial Machine Learning Attacks against Network Intrusion Detection Systems: Classification Analysis

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Abstract – In recent years much research has been focused on the improvement of Network-based Intrusion Detection Systems (NIDS) through the implementation of Machine Learning (ML) approaches. However, together with the numerous assets and promising results from the use of this emerging technology in traditional security systems, novel vulnerabilities arise from the inherent nature of the incorporated ML models. In this paper one such security violation, namely the Adversarial Machine Learning (AML) attack, is studied. While on the one hand, ML can be used to improve the security of the system, conversely it can be also exploited by an adversary, who can benefit from existing ML models to develop more refined attacks in a reduced time and with greater impact. An AML attack is an ML-driven intervention that contaminates the original data for training or testing the ML model of the NIDS, and results in degradation of the performance of the security system. This paper explores the nature of AML attacks and presents a classification analysis of their main types.

Keywords – Adversarial Machine Learning, Network Intrusion Detection Systems, Poisoning attack, Evasion attack.

I. INTRODUCTION

With the evolution of wireless technologies and the upcoming sixth generation (6G) communication systems, a significant development is expected to affect a variety of fields in everyday life including, but not limited to, healthcare, transport, ecology, autonomous manufacturing and diverse other industries. With the growing impact of Machine Learning (ML) applications and Artificial Intelligence (AI), novel services will arise, some of which are associated with massive data generation and exchange. As a consequence, a paradigm shift in cybersecurity issues has emerged that can meet the needs of secure and private systems for data storage, processing and transfer.

Among the applications of ML models in distinct areas, several of the most widely studied are those concerning image-, facial- and speech-recognition, autonomous driving, autonomous disease diagnosis, and large data processing. When cybersecurity is being explored, ML algorithms are most often referred to in threat identification, traffic analysis and anomaly detection, automated incident response and risk assessment. Not only does the introduction of ML enhance the reliability and efficiency of security systems, it also improves their ability to take sophisticated decisions in pattern recognition, with a reduction in the time needed. Moreover, ML enables security systems to deal with evolving threats as well

as those known as zero-day exploits, which are not related to any identification patterns and thus represent a great challenge to conventional security mechanisms.

Despite the diverse benefits introduced by ML applications in the field of cybersecurity, their usage can not be restricted solely to beneficial security aspects. ML can be examined as having a twofold nature, and in the case when it is being used with malicious intent, it can serve as a powerful tool for improving the capabilities of a resourceful attacker. One such ML-based attack, whose negative impact on cybersecurity has attracted research attention during the last few years, consists in the creation of adversarial samples used to train or test ML models and destroy their normal operation. This attack, called Adversarial Machine Learning (AML), is feasible if a small perturbation is induced in the dataset that results in a deceiving outcome from the target ML system [1]. A lot of research is focused on the effect of AML attacks over computer vision and image recognition [2, 3]. However, their impact on traditional security systems, especially for intrusion detection exploiting ML algorithms, can have detrimental consequences on the differentiation of benign and intruded patterns. In this paper the AML attacks that can be launched against Network Intrusion Detection Systems (NIDSs) are studied. A classification of their main types is given according to different categorization parameters.

The rest of the paper is organized as follows: Section II gives a brief overview of ML-based models for NIDS; Section III describes the process of crafting the adversarial sample; in Section IV a classification of AML attacks is presented; Section V concludes the paper.

II. ML MODELS FOR NIDS

Intrusion Detection Systems (IDSs) have a primary role in cybersecurity models, with their ability to recognize unusual traffic or behavior and treat it as being caused by an attacker. The IDS can be established at the user side, which is referred to as host-based IDS, or it can observe the traffic at distinct layers of the network architecture, known as network-based IDS (i.e. NIDS). This paper is focused on NIDSs.

NIDS handles either signature-based detection methods or anomaly-based operational approaches [4]. While the former demands a predefined attack's signature in the database to form a decision, the latter does not suffer from such restrictive requirements and can model the normal network behavior and use it as a reference to discover anomalies. Therefore, the ability of anomaly-based approaches to work adaptively puts them in a more advantageous position in novel attack scenarios when no prior knowledge about the malicious behavior is available.

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The performance of anomaly-based NIDS is further improved by the introduction of ML means in the process of identification between intruded and non-intruded sample. The ML-based NIDS can be implemented by shallow models, which include the classical ML algorithms, and deep learning models with more than one hidden layer in their architecture. Depending on whether the data used in the training phase is categorized in distinct labels or not, the ML models can be classified into supervised learning and unsupervised learning models. In addition there are semi-supervised models where both labeled and unlabeled samples are present in the classifier's training dataset. Among the most popular and frequently used ML models are: Decision Tree, Support Vector Machine, Random Forest, Hidden Markov Models, k-Nearest Neighbors, Naïve Bayes, Genetic Algorithms, Artificial Neural Network, and Logistic Regression. All of them are exhaustively studied in the literature [5, 6].

III. ADVERSARIAL MACHINE LEARNING

In spite of the numerous benefits the introduction of ML algorithms brings to cybersecurity applications, one potential challenge that must be taken into account is their vulnerability to adversarial ML attacks. Adversarial attacks incorporate ML approaches to deceive an ML-based security system about its actual states and make it erroneously classify the outcome. AML can be realized by injecting a small perturbation, that remains indistinguishable, into the original NIDS dataset [7]. Hence, the AML results in an increase of the loss function of the model, thus degrading its performance.

When NIDS is under study, its operation can be described by a binary classifier that distinguishes between two possible states and then labels the samples as intruded or not intruded. In case of AML on such a system, the main goal of the attack is to construct an adversarial sample that has only small variations from the original dataset samples, but still enough to confuse the model and result in prediction errors. Thus, intruded samples can be erroneously labeled not intruded, while benign samples can be classified as malicious. This results in an increase in the false positive and false negative outcomes of the ML model, degrading the detection probability of the NIDS and raising the probability of a false alarm.

The main idea of the formation of adversarial samples is illustrated in Fig. 1 and can be analytically described according to Eqs. (1) and (2) [8]:

$$x' = x + \delta \quad (1)$$

$$f(x) \neq f(x') \quad (2)$$

where x denotes the original set of input data, δ is the perturbation and the resultant adversarial sample is denoted x' . $f(\cdot)$ stands for the classifier, whose outcome in a case of perturbed input data differs from one that is under AML attack.

The impact of AML attacks results in integrity, availability and confidentiality issues concerning both data and systems, hence its nature must be extensively studied.

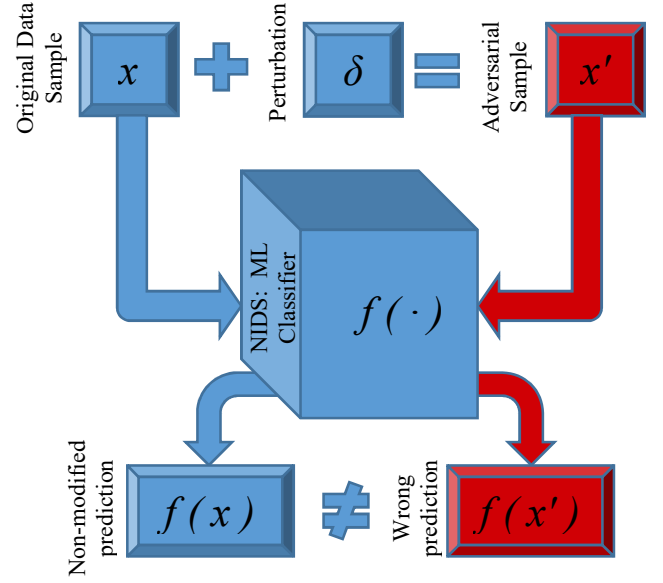


Fig. 1. Generation of adversarial example for AML attack against NIDS

IV. CLASSIFICATION OF AML ATTACKS

AML attacks can be classified according to several different criteria. A systematic enumeration of the main categorizations of AML attacks that can be launched against the security of NIDS is presented in Fig. 2, followed by a brief explanation of each type.

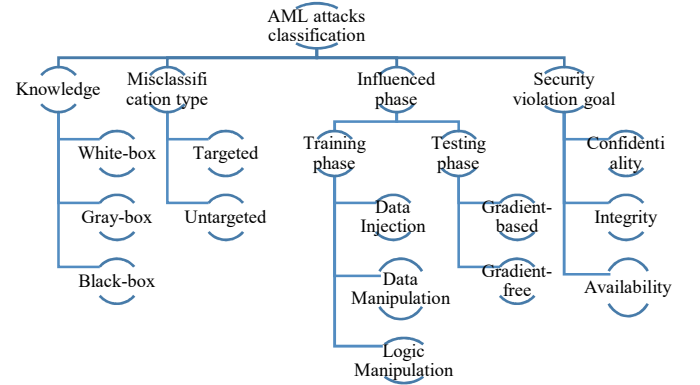


Fig. 2. AML attacks classification

A. Knowledge

The most commonly used criterion to classify the existed AML attacks relates to the extent of knowledge of the intruder about the system. It refers to the ML model architecture incorporated by the NIDS, the learning algorithms, the features and learning parameters used during training, and the training

dataset. The following three different types of malicious activity can be specified [1,7]:

- **White-box attacks:** this is the worst-case scenario, in which the attacker has full information about the target system. Though an intruder having a complete knowledge of the NIDS is not a practical assumption, they make possible the assessment of all the vulnerabilities of the ML system.
- **Gray-box attacks:** these attacking strategies are based on partial knowledge about the target security system, hence they cover a more realistic scenario that can have its practical implementation. In case of a gray-box attack, the adversarial example is generated by taking advantage of a variable extent of information for the ML model or training dataset.
- **Black-box attacks:** the approaches classified as black-box consist in malicious intervention by adversarial example without any knowledge of the ML model or training data of the NIDS. The usual attacking strategy in this case involves querying the target by input data and utilizing its outcomes to produce improved adversarial samples. By iterative algorithms the intruder can expand its knowledge and gradually advance the attack capabilities from being black-box, through gray-box and finally reaching a white-box scenario.

B. Misclassification type

One typical categorization of AML attacks is according to whether the intervention's objective is to produce a particular prediction class or just to confuse the ML model and lead to wrong predictive outcome [7].

- **Targeted:** the attack is targeted at either always predicting a specific class or relating each input data to a particular wrong output.
- **Untargeted:** the adversary aims to deceive the ML model and lead to misclassification with no specific relation between the input data and the misclassified result.

However, in the case of a binary classification model, where only two possible states are available, the targeted misclassification type overlaps with the untargeted one.

C. Influenced phase

Depending on the exact phase of operation of the ML model that is affected, the AML attack can be divided into two main types that appear during the process of training or testing of the model.

- **Training phase:** this type of AML aims at transforming the dataset involved in the training phase of the ML model. It is also known as a poisoning

attack and can be launched by several different approaches, discussed in [9, 10], namely data injection, data manipulation or logic manipulation. While data manipulation with AML adjusts the features or labels of the training data, data injection controls the distribution of data without altering its labels and features. Logic manipulation, on the other hand, is a poisoning attack which tries to corrupt the ML logic rather than the data itself, hence it results in the learning algorithm being controlled by the adversary.

- **Testing phase:** an AML attack that happens during the testing or inference phases seeks for a minor perturbation that leads to erroneous prediction outputs without affecting the ML model. Thus the attack is initiated after the model has already been learned. These types of malicious interventions, called evasion attacks, can be subdivided into gradient-based and gradient-free, depending on whether they rely on any information about the model [1, 10]. Both these forms of evasion attack seek for a solution of an optimization problem whose objective function aims at finding the minimum perturbation that maximizes the loss function of the ML model.

Gradient-based attacks compute a particular distance metric in order to optimize the objective function. They adjust model parameters or features so as to follow the gradient information for the direction of the steepest slope of the objective function where the largest descent or ascent is located. Although the AML attacks that rely on gradient-based methods to obtain the crafted sample are especially useful for finding local extremums, they can give deceptive results when applied to global optimizations. The reason consists in the multimodal and non-linear behavior that can usually be observed in the global optimization functions. As opposed to global optima, local optimization problems are more appropriate to be solved by gradient-based methods, since they can commonly be represented by a smoother outline of the function that changes linearly over time. Moreover, when the optimization function is under the influence of noise, or any other factors destroy its smoothing behavior or cause discontinuities, the use of gradient-based methods for computing the AML example can be unreliable. In these situations, gradient-free optimization methods work more accurately and should be preferred.

Gradient-free AML attacks, on the contrary, do not explore knowledge about the model to optimize the objective function via gradient information. Instead, they employ iterative algorithms to compute the

objective function and discover the optimal perturbation value. Therefore, these methods are valuable tools for the creation of AML examples when no information is available about the model, i.e. in case of black-box attack scenarios. Since the synthesis of the sample requires a large number of attempts, gradient-free evasion attacks are more time-consuming than gradient-based ones. However, due to their independence on the uniform slope of the function, gradient-free methods are applicable for finding global optimal solutions.

D. Security violation goal

According to which security aspect it is trying to undermine, an AML attack may influence the model by three different elements which need to be considered: confidentiality, integrity and availability [1, 10].

- AML against confidentiality: when the attack is targeted against confidentiality, perturbing the model with adversarial examples aims to compromise the privacy of the ML model or its users through successful interception and revealing the data involved in the ML model.
- AML against integrity: the invasion of the ML model through adversarial examples is focused on the integrity of the system such that it results in misclassifications in the model's predictions. This can be realized during either the training phase via poisoning attack or the testing or deployment phase by evasion attack. Depending on the desired misclassification type, both targeted and untargeted attacks can compromise the integrity of the ML model.
- AML against availability: if the security violation goal is to degrade the ML model's availability, the intrusion intends to confuse the model to the extent that it becomes useless. In cases where the effective performance of the predictive model falls below a certain level, the availability attack leads to denial of service of the target system.

V. CONCLUSION

Although ML is widely used to improve the security of contemporary networks and increase the potential of NIDS to discover malicious activities, it can also be adopted for malware activities. One such security violation explores the power of ML algorithms to craft an adversarial sample and then use it to deceive the model of NIDS and destroy its normal operation. As a result, the loss function of the model increases, its effective performance is degraded, and subsequent

intrusions may successfully pass through the protection of the NIDS.

In this paper the influence of AML attacks against the security of NIDSs is studied. A classification of the main types of this attack is presented according to four different categorization parameters: level of knowledge about the legitimate system that is available at the adversary; target of produced misclassification; phase of the model that is under influence; security violation goal. All types of AML attacks are briefly described, together with their most significant subtypes. This study can serve as a primal direction of a future research aimed at comparative analyses of the most significant methods for AML attacks, defense opportunities against them and novel contributions in these topics that are expected to have a severe impact on the emerging topic of cybersecurity.

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